Scale Exercises - Section IX, Level 71

A melodic minor intervals, chord overlays, functional modes, and progression-specific applications

FIXED KEY: A | 10 MODULES | 800 TONES

Level 70 connected the A-melodic-minor fretboard. Level 71 reorganizes that map by interval, chord, mode, and harmonic function. The mode name is useful only when the active chord and destination are understood.

1. Hear the interval quality

The same diatonic pattern produces major, minor, perfect, augmented, and diminished intervals inside melodic minor.

2. Overlay the harmony

Triads and seventh chords remain the structural skeleton beneath every mode and melodic phrase.

3. Name function first

D Lydian dominant, E Mixolydian b6, and G# altered are chord-scale functions, not random exotic box names.

A melodic minor harmony family

```
A melodic minor  A B C D E F# G#      1 2 b3 4 5 6 7

TRIADS
i Am | ii Bm | III+ Caug | IV D | V E | vi-dim F#dim | vii-dim G#dim

SEVENTH CHORDS
iMaj7 AmMaj7 | ii7 Bm7 | IIImaj7#5 Cmaj7#5 | IV7 D7
V7 E7 | vi-half-dim F#m7b5 | vii-half-dim G#m7b5

MODE-7 DUAL READING
Parent stack: G#m7b5 -> AmMaj7
Applied dominant respelling: G#7alt -> C# minor
```

Written workload

Module	Tones
A	240
B	56
C	56
D	112
E	56
F	80
G	48
H	40
I	48
J	64
TOTAL	800

Session protocol

Choose one interval family, one chord overlay, and one mode-function application. State the active chord before the mode. Use clean tone, preserve the beat-one target, and remove notes whenever the harmonic function becomes unclear.

MODE RULE A mode formula describes the local root. The parent A-melodic-minor degrees remain the same seven pitches.

Scale Exercises - Section IX, Level 71

Mode/function atlas and seventh-mode dual interpretation

CHORD FIRST | MODE SECOND | DESTINATION THIRD

SEVEN MODES SHARE ONE PITCH COLLECTION BUT DO NOT SHARE ONE FUNCTION

Degree	Mode	Formula	Primary chord function
1	A melodic minor	1 2 b3 4 5 6 7	AmMaj7 / iMaj7
2	B Dorian b2	1 b2 b3 4 5 6 b7	Bm7 / ii7
3	C Lydian augmented	1 2 3 #4 #5 6 7	Cmaj7#5 / IIImaj7#5
4	D Lydian dominant	1 2 3 #4 5 6 b7	D7#11 / IV7
5	E Mixolydian b6	1 2 3 4 5 b6 b7	E7b13 / V7
6	F# Locrian natural 2	1 2 b3 4 b5 b6 b7	F#m7b5 / vi-half-dim

Parent-key reading

Stacking diatonic thirds on scale degree 7 produces G#-B-D-F#: G#m7b5. In the parent A-minor system, it can resolve by half-step motion into AmMaj7.

Applied altered-dominant reading

When G# is treated as a dominant root, C is respelled B# as the major third. The same pitch collection becomes G# altered and points toward C# major or minor.

PRACTICE ORDER Chord arpeggio -> one-octave mode -> characteristic note -> destination chord.

WARNING Do not apply altered-scale tension to G#m7b5 unless the harmony has actually been reinterpreted as G#7alt.

Scale Exercises - Section IX, Level 71

Module A - P1 tonic frame interval sequences

P1 | 48 TONES

Diatonic thirds

Six ascending interval pairs; 12 picked tones.

```
Nte|A---C---B---D---C---E---D---F#--E---G#--F#--A---
Deg|1---b3--2---4---b3--5---4---6---5---7---6---1---
  D|-----------------------------------6-------7---
  A|-----------5-------7---5---9---7-------9-------
  E|5---8---7-------8-----------------------------
```

Diatonic fourths

Six ascending interval pairs; 12 picked tones.

```
Nte|A---D---B---E---C---F#--D---G#--E---A---F#--B---
Deg|1---4---2---5---b3--6---4---7---5---1---6---2---
  D|---------------------------6-------7-------9---
  A|----5-------7-------9---5-------7-------9-------
  E|5-------7-------8-----------------------------
```

Diatonic fifths

Six ascending interval pairs; 12 picked tones.

```
Nte|A---E---B---F#--C---G#--D---A---E---B---F#--C---
Deg|1---5---2---6---b3--7---4---1---5---2---6---b3--
  G|-----------------------------------------5---
  D|-------------------6-------7-------9-----------
  A|----7-------9-----------5-------7-------9-------
  E|5-------7-------8-----------------------------
```

Diatonic sixths

Six ascending interval pairs; 12 picked tones.

```
Nte|A---F#--B---G#--C---A---D---B---E---C---F#--D---
Deg|1---6---2---7---b3--1---4---2---5---b3--6---4---
  G|-----------------------------------5-------7---
  D|-----------6-------7-------9-----------------
  A|----9-----------------5-------7-------9-------
  E|5-------7-------8-----------------------------
```

Standard tuning. Fixed key: A melodic minor. Modes are tied to chord function, not treated as unrelated scale boxes.

Scale Exercises - Section IX, Level 71

Module A - P2 second-degree frame interval sequences

P2 | 48 TONES

Diatonic thirds

Six ascending interval pairs; 12 picked tones.

```
Nte|B---D---C---E---D---F#--E---G#--F#--A---G#--B---
Deg|2---4---b3--5---4---6---5---7---6---1---7---2---
  D|-----------------------------------7-------9---
  A|-----------7-------9---7---11--9-------11------
  E|7---10--8-------10----------------------------
```

Diatonic fourths

Six ascending interval pairs; 12 picked tones.

```
Nte|B---E---C---F#--D---G#--E---A---F#--B---G#--C---
Deg|2---5---b3--6---4---7---5---1---6---2---7---b3--
  D|---------------------------7-------9-------10--
  A|----7-------9-------11--7-------9-------11------
  E|7-------8-------10----------------------------
```

Diatonic fifths

Six ascending interval pairs; 12 picked tones.

```
Nte|B---F#--C---G#--D---A---E---B---F#--C---G#--D---
Deg|2---6---b3--7---4---1---5---2---6---b3--7---4---
  G|-------------------------------------------7---
  D|-------------------7-------9-------10----------
  A|----9-------11----------7-------9-------11------
  E|7-------8-------10----------------------------
```

Diatonic sixths

Six ascending interval pairs; 12 picked tones.

```
Nte|B---G#--C---A---D---B---E---C---F#--D---G#--E---
Deg|2---7---b3--1---4---2---5---b3--6---4---7---5---
  G|-----------------------------------7-------9---
  D|-----------7-------9-------10------------------
  A|----11------------------7-------9-------11------
  E|7-------8-------10----------------------------
```

Standard tuning. Fixed key: A melodic minor. Modes are tied to chord function, not treated as unrelated scale boxes.

Scale Exercises - Section IX, Level 71

Module A - P3 flat-third frame interval sequences

P3 | 48 TONES

Diatonic thirds

Six ascending interval pairs; 12 picked tones.

```
Nte|C---E---D---F#--E---G#--F#--A---G#--B---A---C---
Deg|b3--5---4---6---5---7---6---1---7---2---1---b3--
  D|-------------------------------------9-------10--
  A|-----------9-------11--9---12--11------12------
  E|8---12--10------12----------------------------
```

Diatonic fourths

Six ascending interval pairs; 12 picked tones.

```
Nte|C---F#--D---G#--E---A---F#--B---G#--C---A---D---
Deg|b3--6---4---7---5---1---6---2---7---b3--1---4---
  D|---------------------------9-------10------12--
  A|----9-------11------12--9-------11------12------
  E|8-------10------12----------------------------
```

Diatonic fifths

Six ascending interval pairs; 12 picked tones.

```
Nte|C---G#--D---A---E---B---F#--C---G#--D---A---E---
Deg|b3--7---4---1---5---2---6---b3--7---4---1---5---
  G|-----------------------------------------9---
  D|-------------------9-------10------12----------
  A|----11------12----------9-------11------12------
  E|8-------10------12----------------------------
```

Diatonic sixths

Six ascending interval pairs; 12 picked tones.

```
Nte|C---A---D---B---E---C---F#--D---G#--E---A---F#--
Deg|b3--1---4---2---5---b3--6---4---7---5---1---6---
  G|-----------------------------------9-------11--
  D|-----------9-------10------12------------------
  A|----12-----------------9-------11------12------
  E|8-------10------12----------------------------
```

Standard tuning. Fixed key: A melodic minor. Modes are tied to chord function, not treated as unrelated scale boxes.

Scale Exercises - Section IX, Level 71

Module A - P4 fourth-degree frame interval sequences

P4 | 48 TONES

Diatonic thirds

Six ascending interval pairs; 12 picked tones.

```
Nte|D---F#--E---G#--F#--A---G#--B---A---C---B---D---
Deg|4---6---5---7---6---1---7---2---1---b3--2---4---
  D|-----------------------------------10------12--
  A|-----------11------12--11--14--12------14------
  E|10--14--12------14----------------------------
```

Diatonic fourths

Six ascending interval pairs; 12 picked tones.

```
Nte|D---G#--E---A---F#--B---G#--C---A---D---B---E---
Deg|4---7---5---1---6---2---7---b3--1---4---2---5---
  D|---------------------------10------12------14--
  A|----11------12------14--11------12------14------
  E|10------12------14----------------------------
```

Diatonic fifths

Six ascending interval pairs; 12 picked tones.

```
Nte|D---A---E---B---F#--C---G#--D---A---E---B---F#--
Deg|4---1---5---2---6---b3--7---4---1---5---2---6---
  G|-----------------------------------------------11--
  D|-------------------10------12------14----------
  A|----12------14----------11------12------14------
  E|10------12------14----------------------------
```

Diatonic sixths

Six ascending interval pairs; 12 picked tones.

```
Nte|D---B---E---C---F#--D---G#--E---A---F#--B---G#--
Deg|4---2---5---b3--6---4---7---5---1---6---2---7---
  G|-----------------------------------11------13--
  D|-----------10------12------14-----------------
  A|----14-----------------11------12------14------
  E|10------12------14----------------------------
```

Standard tuning. Fixed key: A melodic minor. Modes are tied to chord function, not treated as unrelated scale boxes.

Scale Exercises - Section IX, Level 71

Module A - P5 fifth-degree frame interval sequences

P5 | 48 TONES

Diatonic thirds

Six ascending interval pairs; 12 picked tones.

```
Nte|E---G#--F#--A---G#--B---A---C---B---D---C---E---
Deg|5---7---6---1---7---2---1---b3--2---4---b3--5---
  D|-----------------------------------12------14--
  A|-----------12------14--12--15--14------15------
  E|12--16--14------16-----------------------------
```

Diatonic fourths

Six ascending interval pairs; 12 picked tones.

```
Nte|E---A---F#--B---G#--C---A---D---B---E---C---F#--
Deg|5---1---6---2---7---b3--1---4---2---5---b3--6---
  D|---------------------------12------14------16--
  A|----12------14------15--12------14------15------
  E|12------14------16-----------------------------
```

Diatonic fifths

Six ascending interval pairs; 12 picked tones.

```
Nte|E---B---F#--C---G#--D---A---E---B---F#--C---G#--
Deg|5---2---6---b3--7---4---1---5---2---6---b3--7---
  G|-----------------------------------------13--
  D|-------------------12------14------16----------
  A|----14------15----------12------14------15------
  E|12------14------16-----------------------------
```

Diatonic sixths

Six ascending interval pairs; 12 picked tones.

```
Nte|E---C---F#--D---G#--E---A---F#--B---G#--C---A---
Deg|5---b3--6---4---7---5---1---6---2---7---b3--1---
  G|-----------------------------------13------14--
  D|-----------12------14------16------------------
  A|----15-----------------12------14------15------
  E|12------14------16-----------------------------
```

Standard tuning. Fixed key: A melodic minor. Modes are tied to chord function, not treated as unrelated scale boxes.

Scale Exercises - Section IX, Level 71

Module B - harmonized-triad arpeggio overlays

7 TRIADS | 56 TONES

ROOT-THIRD-FIFTH-OCTAVE; RE-ATTACK THE TOP ROOT AND REVERSE

The raised sixth changes Bdim to Bm and Dm to D. The augmented C chord and both diminished triads remain essential parts of the parent harmony.

Am

i - Am: A-C-E

```
Nte|A----C----E----A----A----E----C----A----
Deg|1----b3---5----1----1----5----b3---1----
  D|---------------7----7------------------
  A|----------7--------------7--------------
  E|5----8-----------------------8----5----
```

Bm

ii - Bm: B-D-F#

```
Nte|B----D----F#---B----B----F#---D----B----
Deg|2----4----6----2----2----6----4----2----
  D|---------------9----9------------------
  A|-----5----9--------------9----5---------
  E|7-----------------------------------7----
```

Caug

III+ - Caug: C-E-G#

```
Nte|C----E----G#---C----C----G#---E----C----
Deg|b3---5----7----b3---b3---7----5----b3---
  G|---------------5----5------------------
  D|----------6--------------6--------------
  A|-----7-----------------------7---------
  E|8-----------------------------------8----
```

D

IV - D: D-F#-A

```
Nte|D----F#---A----D----D----A----F#---D----
Deg|4----6----1----4----4----1----6----4----
  G|---------------7----7------------------
  D|----------7--------------7--------------
  A|5----9-----------------------9----5----
```

E

V - E: E-G#-B

```
Nte|E----G#---B----E----E----B----G#---E----
Deg|5----7----2----5----5----2----7----5----
  G|---------------9----9------------------
  D|-----6----9--------------9----6---------
  A|7-----------------------------------7----
```

F#dim

vi-dim - F#dim: F#-A-C

```
Nte|F#---A----C----F#---F#---C----A----F#---
Deg|6----1----b3---6----6----b3---1----6----
  B|---------------7----7------------------
  G|----------5--------------5--------------
  D|-----7-----------------------7---------
  A|9-----------------------------------9----
```

G#dim

vii-dim - G#dim: G#-B-D

```
Nte|G#---B----D----G#---G#---D----B----G#---
Deg|7----2----4----7----7----4----2----7----
  B|---------------9----9------------------
  G|----------7--------------7--------------
  D|6----9-----------------------9----6----
```

Scale Exercises - Section IX, Level 71

Module C - harmonized seventh-chord arpeggio overlays

7 CHORDS | 56 TONES

ROOT-THIRD-FIFTH-SEVENTH; QUALITY-DEFINING TONES MUST REMAIN AUDIBLE

Accent the third and seventh on the first pass. On half-diminished chords, also accent the flat fifth. The mode-function work on the next pages begins from these chord identities.

AmMaj7

iMaj7 - AmMaj7: A-C-E-G#

```
Nte|A----C----E----G#---G#---E----C----A----
Deg|1----b3---5----7----7----5----b3---1----
  D|---------------6----6------------------
  A|----------7--------------7--------------
  E|5----8-----------------------8----5----
```

Bm7

ii7 - Bm7: B-D-F#-A

```
Nte|B----D----F#---A----A----F#---D----B----
Deg|2----4----6----1----1----6----4----2----
  D|---------------7----7------------------
  A|-----5----9--------------9----5---------
  E|7----------------------------------7----
```

Cmaj7#5

IIImaj7#5 - Cmaj7#5: C-E-G#-B

```
Nte|C----E----G#---B----B----G#---E----C----
Deg|b3---5----7----2----2----7----5----b3---
  D|----------6----9----9----6--------------
  A|-----7-----------------------7----------
  E|8----------------------------------8----
```

D7

IV7 - D7: D-F#-A-C

```
Nte|D----F#---A----C----C----A----F#---D----
Deg|4----6----1----b3---b3---1----6----4----
  G|--------------5----5------------------
  D|----------7--------------7--------------
  A|5----9-----------------------9----5----
```

E7

V7 - E7: E-G#-B-D

```
Nte|E----G#---B----D----D----B----G#---E----
Deg|5----7----2----4----4----2----7----5----
  G|---------------7----7------------------
  D|-----6----9--------------9----6---------
  A|7----------------------------------7----
```

F#m7b5

vi-half-dim - F#m7b5: F#-A-C-E

```
Nte|F#---A----C----E----E----C----A----F#---
Deg|6----1----b3---5----5----b3---1----6----
  G|----------5----9----9----5--------------
  D|-----7-----------------------7----------
  A|9----------------------------------9----
```

G#m7b5

vii-half-dim - G#m7b5: G#-B-D-F#

```
Nte|G#---B----D----F#---F#---D----B----G#---
Deg|7----2----4----6----6----4----2----7----
  B|---------------7----7------------------
  G|----------7--------------7--------------
  D|6----9-----------------------9----6----
```

Scale Exercises - Section IX, Level 71

Module D - one-octave functional mode mirrors

MODES 1-4 | 64 TONES

SUSTAIN THE ASSOCIATED CHORD OR DRONE; SPEAK THE MODE-RELATIVE FORMULA

Play one-octave ascent, re-attack the upper root, and reverse exactly. The parent notes remain A-B-C-D-E-F#-G#, but the relative formula changes with the local root.

A melodic minor

1 2 b3 4 5 6 7 1 | AmMaj7 / tonic minor-major

```
Nte|A---B---C---D---E---F#--G#--A---A---G#--F#--E---D---C---B---A---
Deg|1---2---b3--4---5---6---7---1---1---7---6---5---4---b3--2---1---
  D|------------------------6---7---7---6---------------------------
  A|-----------5---7---9-------------------9---7---5---------------
  E|5---7---8-----------------------------------------8---7---5---
```

B Dorian b2

1 b2 b3 4 5 6 b7 1 | Bm7 / minor predominant color

```
Nte|B---C---D---E---F#--G#--A---B---B---A---G#--F#--E---D---C---B---
Deg|2---b3--4---5---6---7---1---2---2---1---7---6---5---4---b3--2---
  D|------------------------7---9---9---7---------------------------
  A|-----------7---9---11------------------11--9---7---------------
  E|7---8---10-----------------------------------------10--8---7---
```

C Lydian augmented

1 2 3 #4 #5 6 7 1 | Cmaj7#5 / augmented-major color

```
Nte|C---D---E---F#--G#--A---B---C---C---B---A---G#--F#--E---D---C---
Deg|b3--4---5---6---7---1---2---b3--b3--2---1---7---6---5---4---b3--
  D|------------------------9---10--10--9---------------------------
  A|-----------9---11--12------------------12--11--9---------------
  E|8---10--12-----------------------------------------12--10--8---
```

D Lydian dominant

1 2 3 #4 5 6 b7 1 | D7#11 / dominant with Lydian color

```
Nte|D---E---F#--G#--A---B---C---D---D---C---B---A---G#--F#--E---D---
Deg|4---5---6---7---1---2---b3--4---4---b3--2---1---7---6---5---4---
  D|------------------------10--12--12--10--------------------------
  A|-----------11--12--14------------------14--12--11--------------
  E|10--12--14-----------------------------------------14--12--10--
```

Scale Exercises - Section IX, Level 71

Module D - one-octave functional mode mirrors

MODES 5-7 | 48 TONES

SUSTAIN THE ASSOCIATED CHORD OR DRONE; SPEAK THE MODE-RELATIVE FORMULA

Play one-octave ascent, re-attack the upper root, and reverse exactly. The parent notes remain A-B-C-D-E-F#-G#, but the relative formula changes with the local root.

E Mixolydian b6

1 2 3 4 5 b6 b7 1 | E7b13 / dominant with minor-sixth color

```
Nte|E---F#--G#--A---B---C---D---E---E---D---C---B---A---G#--F#--E---
Deg|5---6---7---1---2---b3--4---5---5---4---b3--2---1---7---6---5---
  D|------------------------12--14--14--12--------------------------
  A|------------12--14--15-----------------15--14--12--------------
  E|12--14--16-----------------------------------16--14--12--
```

F# Locrian natural 2

1 2 b3 4 b5 b6 b7 1 | F#m7b5 / half-diminished function

```
Nte|F#--G#--A---B---C---D---E---F#--F#--E---D---C---B---A---G#--F#--
Deg|6---7---1---2---b3--4---5---6---6---5---4---b3--2---1---7---6---
  D|--------------------12--14--16--16--14--12----------------------
  A|--------12--14--15--------------------------15--14--12----------
  E|14--16-------------------------------------------------16--14--
```

G# altered

1 b9 #9 3 b5 #5 b7 1 | G#7alt applied dominant; parent diatonic stack G#m7b5

```
Nte|G#--A---B---C---D---E---F#--G#--G#--F#--E---D---C---B---A---G#--
Deg|7---1---2---b3--4---5---6---7---7---6---5---4---b3--2---1---7---
  G|----------------------------13--13----------------------------
  D|----------------12--14--16----------16--14--12-----------------
  A|----12--14--15---------------------------------15--14--12------
  E|16-----------------------------------------------------------16--
```

G# altered has no natural fifth. Hear R-3-b7 first; add b9, #9, b5, or #5 only after the dominant function is clear.

Scale Exercises - Section IX, Level 71

Module E - mode-function melodic cells

7 CELLS | 56 TONES

BEGIN ON A CHORD TONE, TOUCH THE CHARACTERISTIC COLOR, AND RETURN TO THE FUNCTIONAL CHORD

These are not scale runs. Each eight-tone cell is a short melodic statement designed to expose the chord quality and one characteristic mode color.

A melodic minor

AmMaj7 / tonic minor-major

```
Nte|A----C----E----G#---F#---E----C----A----
Deg|1----b3---5----7----6----5----b3---1----
  D|--------------6-----------------------
  A|----------7---------9----7-------------
  E|5----8-----------------------8----5----
```

B Dorian b2

Bm7 / minor predominant color

```
Nte|B----D----F#---A----C----G#---F#---B----
Deg|2----4----6----1----b3---7----6----2----
  D|--------------7-----------------------
  A|----------9-------------11---9--------
  E|7----10------------8-------------7----
```

C Lydian augmented

Cmaj7#5 / augmented-major color

```
Nte|C----E----G#---B----F#---G#---E----C----
Deg|b3---5----7----2----6----7----5----b3---
  D|----------6----9---------6--------------
  A|-----7-------------9---------7---------
  E|8---------------------------------8----
```

D Lydian dominant

D7#11 / dominant with Lydian color

```
Nte|D----F#---A----C----G#---A----F#---D----
Deg|4----6----1----b3---7----1----6----4----
  G|--------------5-----------------------
  D|----------7---------6----7--------------
  A|5----9-----------------------9----5----
```

E Mixolydian b6

E7b13 / dominant with minor-sixth color

```
Nte|E----G#---B----D----C----B----G#---E----
Deg|5----7----2----4----b3---2----7----5----
  D|--------------12----------------------
  A|-----11---14--------15---14---11-------
  E|12--------------------------------12---
```

F# Locrian natural 2

F#m7b5 / half-diminished function

```
Nte|F#---A----C----E----G#---E----C----F#---
Deg|6----1----b3---5----7----5----b3---6----
  D|--------------14--------14-------------
  A|-----12---15--------11--------15-------
  E|14--------------------------------14---
```

G# altered

G#7alt applied dominant; parent diatonic stack G#m7b5

```
Nte|G#---C----F#---A----B----D----E----G#---
Deg|7----b3---6----1----2----4----5----7----
  B|----------7-----------------------------
  G|-----5-----------------------------------
  D|-----------------------12---14--------
  A|--------------12---14-----------------
  E|16--------------------------------16---
```

Scale Exercises - Section IX, Level 71

Module F - five-position iMaj7-IV7-V7-iMaj7 melodic phrases

P1-P3 | 48 TONES

| AmMaj7 | D7 | E7 | AmMaj7 | - BEAT 1 IS A CHORD TONE

The IV chord is major or dominant because melodic minor contains F#. The final G# of E7 resolves into A or remains as the major seventh of AmMaj7, depending on the printed destination.

P1 - 16-tone cadence phrase

Target first; rephrase notes 2-4 only after the quarter-note pass is secure.

```
Nte|A---C---E---G#--|D---F#--A---C---|E---G#--B---D---|A---C---E---G#--|
Deg|1---b3--5---7---|4---6---1---b3--|5---7---2---4---|1---b3--5---7---|
Rol|>R--b3--5---7---|>R--3---5---b7--|>R--3---5---b7--|>R--b3--5---7---|
  B|----------------|----------------|----------------|------------9---|
  G|----------------|------------5---|------------7---|----5---9-------|
  D|------------6---|--------7-------|----6---9-------|7---------------|
  A|--------7-------|5---9-----------|7---------------|----------------|
  E|5---8-----------|----------------|----------------|----------------|
```

P2 - 16-tone cadence phrase

Target first; rephrase notes 2-4 only after the quarter-note pass is secure.

```
Nte|A---C---E---G#--|D---F#--A---C---|E---G#--B---D---|A---C---E---G#--|
Deg|1---b3--5---7---|4---6---1---b3--|5---7---2---4---|1---b3--5---7---|
Rol|>R--b3--5---7---|>R--3---5---b7--|>R--3---5---b7--|>R--b3--5---7---|
  B|------------9---|----------------|----------------|------------9---|
  G|--------9-------|----------------|------------7---|--------9-------|
  D|7---10----------|--------7---10--|--------9-------|7---10----------|
  A|----------------|----9-----------|7---11----------|----------------|
  E|----------------|10--------------|----------------|----------------|
```

P3 - 16-tone cadence phrase

Target first; rephrase notes 2-4 only after the quarter-note pass is secure.

```
Nte|A---C---E---G#--|D---F#--A---C---|E---G#--B---D---|A---C---E---G#--|
Deg|1---b3--5---7---|4---6---1---b3--|5---7---2---4---|1---b3--5---7---|
Rol|>R--b3--5---7---|>R--3---5---b7--|>R--3---5---b7--|>R--b3--5---7---|
  G|--------9---13--|----------------|----------------|--------9---13--|
  D|----10----------|------------10--|--------9---12--|----10----------|
  A|12--------------|----9---12------|----11----------|12--------------|
  E|----------------|10--------------|12--------------|----------------|
```

Scale Exercises - Section IX, Level 71

Module F - five-position iMaj7-IV7-V7-iMaj7 melodic phrases

P4-P5 | 32 TONES

| AmMaj7 | D7 | E7 | AmMaj7 | - BEAT 1 IS A CHORD TONE

The IV chord is major or dominant because melodic minor contains F#. The final G# of E7 resolves into A or remains as the major seventh of AmMaj7, depending on the printed destination.

P4 - 16-tone cadence phrase

Target first; rephrase notes 2-4 only after the quarter-note pass is secure.

```
Nte|A---C---E---G#--|D---F#--A---C---|E---G#--B---D---|A---C---E---G#--|
Deg|1---b3--5---7---|4---6---1---b3--|5---7---2---4---|1---b3--5---7---|
Rol|>R--b3--5---7---|>R--3---5---b7--|>R--3---5---b7--|>R--b3--5---7---|
  G|------------13--|----------------|----------------|------------13--|
  D|----10--14------|------------10--|------------12--|----10--14------|
  A|12--------------|--------12------|----11--14------|12--------------|
  E|----------------|10--14----------|12--------------|----------------|
```

P5 - 16-tone cadence phrase

Target first; rephrase notes 2-4 only after the quarter-note pass is secure.

```
Nte|A---C---E---G#--|D---F#--A---C---|E---G#--B---D---|A---C---E---G#--|
Deg|1---b3--5---7---|4---6---1---b3--|5---7---2---4---|1---b3--5---7---|
Rol|>R--b3--5---7---|>R--3---5---b7--|>R--3---5---b7--|>R--b3--5---7---|
  B|----------------|------------13--|----------------|----------------|
  G|------------13--|--------14------|----------------|------------13--|
  D|--------14------|12--16----------|------------12--|--------14------|
  A|12--15----------|----------------|--------14------|12--15----------|
  E|----------------|----------------|12--16----------|----------------|
```

Variation: end the final Am chord on F# for Am6 color, then repeat and end on G# for AmMaj7 color.

Scale Exercises - Section IX, Level 71

Module G - alternate melodic-minor progression families

3 PHRASES | 48 TONES

THREE HARMONIC PATHS; FOUR CHORD TONES PER BAR

The phrases establish the new ii7 and IV7 qualities, the half-diminished vi chord, and the augmented-major III chord. Every measure begins on a chord tone.

ii7-V7-iMaj7-i6

Four bars; four chord tones per bar.

```
Nte|B---D---F#--A---|E---G#--B---D---|A---C---E---G#--|A---C---E---F#--|
Deg|2---4---6---1---|5---7---2---4---|1---b3--5---7---|1---b3--5---6---|
Rol|>R--b3--5---b7--|>R--3---5---b7--|>R--b3--5---7---|>R--b3--5---6---|
  G|----------------|------------7---|----------------|----------------|
  D|------------7---|----6---9-------|------------6---|----------------|
  A|----5---9-------|7---------------|--------7-------|--------7---9---|
  E|7---------------|----------------|5---8-----------|5---8-----------|
```

vi-half-dim-ii7-V7-iMaj7

Four bars; four chord tones per bar.

```
Nte|F#--A---C---E---|B---D---F#--A---|E---G#--B---D---|A---C---E---G#--|
Deg|6---1---b3--5---|2---4---6---1---|5---7---2---4---|1---b3--5---7---|
Rol|>R--b3--b5--b7--|>R--b3--5---b7--|>R--3---5---b7--|>R--b3--5---7---|
  G|--------5---9---|----------------|------------7---|----------------|
  D|----7-----------|------------7---|----6---9-------|------------6---|
  A|9---------------|----5---9-------|7---------------|--------7-------|
  E|----------------|7---------------|----------------|5---8-----------|
```

IIImaj7#5-IV7-V7-iMaj7

Four bars; four chord tones per bar.

```
Nte|C---E---G#--B---|D---F#--A---C---|E---G#--B---D---|A---C---E---G#--|
Deg|b3--5---7---2---|4---6---1---b3--|5---7---2---4---|1---b3--5---7---|
Rol|>R--3---#5--7---|>R--3---5---b7--|>R--3---5---b7--|>R--b3--5---7---|
  G|----------------|------------5---|------------7---|----------------|
  D|--------6---9---|--------7-------|----6---9-------|------------6---|
  A|----7-----------|5---9-----------|7---------------|--------7-------|
  E|8---------------|----------------|----------------|5---8-----------|
```

Standard tuning. Fixed key: A melodic minor. Modes are tied to chord function, not treated as unrelated scale boxes.

Scale Exercises - Section IX, Level 71

Module H - D Lydian dominant through five positions

5 CELLS | 40 TONES

D-E-F#-G#-A-B-C-D | 1-2-3-#4-5-6-b7-1

The raised fourth G# separates D Lydian dominant from ordinary Mixolydian. Keep F# and C audible so the phrase still identifies D7 while the #4 adds the characteristic brightness.

P1

D7#11 function; 8 picked tones.

```
Nte|D----E----F#---G#---A----B----C----D----
Deg|4----5----6----7----1----2----b3---4----
  G|-----------------------------5----7----
  D|--------------6----7----9--------------
  A|5----7----9----------------------------
```

P2

D7#11 function; 8 picked tones.

```
Nte|D----E----F#---G#---A----B----C----D----
Deg|4----5----6----7----1----2----b3---4----
  G|-----------------------------------7----
  D|-------------------7----9----10--------
  A|-----7----9----11-----------------------
  E|10-------------------------------------
```

P3

D7#11 function; 8 picked tones.

```
Nte|D----E----F#---G#---A----B----C----D----
Deg|4----5----6----7----1----2----b3---4----
  D|-----------------------9----10---12---
  A|----------9----11---12-----------------
  E|10---12---------------------------------
```

P4

D7#11 function; 8 picked tones.

```
Nte|D----E----F#---G#---A----B----C----D----
Deg|4----5----6----7----1----2----b3---4----
  D|----------------------------10---12---
  A|---------------11---12---14-------------
  E|10---12---14--------------------------
```

P5

D7#11 function; 8 picked tones.

```
Nte|D----E----F#---G#---A----B----C----D----
Deg|4----5----6----7----1----2----b3---4----
  B|----------------------------13---15---
  G|---------------13---14---16-------------
  D|12---14---16--------------------------
```

Chord-tone emphasis

First accent D-F#-A-C. Then accent G# alone. Finally remove all accents while keeping the D7 identity and the #4 color audible.

Scale Exercises - Section IX, Level 71

Module I - characteristic resolutions and mode-7 dual interpretation

6 CELLS | 48 TONES

FOUR-TONE SOURCE CHORD -> FOUR-TONE DESTINATION CHORD

Play the source as an arpeggio, leave one eighth-note rest, and play the destination. Track the smallest movement between voices before increasing tempo.

AmMaj7 -> Am6

Source | destination

```
Nte|A---C---E---G#--|A---C---E---F#--
Deg|1---b3--5---7---|1---b3--5---6---
Rol|R---b3--5---7---|R---b3--5---6---
  D|-----------6---|---------------
  A|--------7-------|--------7---9---
  E|5---8-----------|5---8-----------
```

Cmaj7#5 -> AmMaj7

Source | destination

```
Nte|C---E---G#--B---|A---C---E---G#--
Deg|b3--5---7---2---|1---b3--5---7---
Rol|R---3---#5--7---|R---b3--5---7---
  D|--------6---9---|-----------6---
  A|----7-----------|--------7-------
  E|8---------------|5---8-----------
```

Bm7 -> E7

Source | destination

```
Nte|B---D---F#--A---|E---G#--B---D---
Deg|2---4---6---1---|5---7---2---4---
Rol|R---b3--5---b7--|R---3---5---b7--
  G|---------------|-----------7---
  D|-----------7---|----6---9-------
  A|----5---9-------|7---------------
  E|7---------------|----------------
```

F#m7b5 -> E7

Source | destination

```
Nte|F#--A---C---E---|E---G#--B---D---
Deg|6---1---b3--5---|5---7---2---4---
Rol|R---b3--b5--b7--|R---3---5---b7--
  G|--------5---9---|-----------7---
  D|----7-----------|----6---9-------
  A|9---------------|7---------------
```

G#m7b5 -> AmMaj7

Source | destination

```
Nte|G#--B---D---F#--|A---C---E---G#--
Deg|7---2---4---6---|1---b3--5---7---
Rol|R---b3--b5--b7--|R---b3--5---7---
  B|-----------7---|-----------9---
  G|--------7-------|----5---9-------
  D|6---9-----------|7---------------
```

G#7alt -> C#m7

Source | destination

```
Nte|G#--C---F#--A---|C#--E---G#--B---
Deg|7---b3--6---1---|R*--5---7---2---
Rol|R---3---b7--b9--|R---b3--5---b7--
  B|--------7---10--|--------9---12--
  G|----5-----------|6---9-----------
  D|6---------------|----------------
```

Mode-7 naming rule

G#m7b5 belongs to the parent-key stack and resolves toward A minor. G#7alt is an applied dominant interpretation and resolves toward C#. Use the same notes differently only when the harmony and destination justify the respelling.

Scale Exercises - Section IX, Level 71

Module J - sixteen-bar mode-function capstone, bars 1-8

32 TONES | FIRST HALF

FOUR CHORD TONES PER BAR; ANNOUNCE THE MODE OR CHORD FUNCTION BEFORE BEAT 1

The first eight bars rotate through the complete parent harmony. The second half establishes functional cadences and tonic colors. Later passes may add one characteristic mode tone after the beat-one target.

Bar 1 - AmMaj7

Four quarter-note chord tones.

```
Nte|A---C---E---G#--
Deg|1---b3--5---7---
Rol|>R--b3--5---7---
  D|------------6---
  A|--------7-------
  E|5---8-----------
```

Bar 2 - Bm7

Four quarter-note chord tones.

```
Nte|B---D---F#--A---
Deg|2---4---6---1---
Rol|>R--b3--5---b7--
  D|------------7---
  A|----5---9-------
  E|7---------------
```

Bar 3 - Cmaj7#5

Four quarter-note chord tones.

```
Nte|C---E---G#--B---
Deg|b3--5---7---2---
Rol|>R--3---#5--7---
  D|--------6---9---
  A|----7-----------
  E|8---------------
```

Bar 4 - D7

Four quarter-note chord tones.

```
Nte|D---F#--A---C---
Deg|4---6---1---b3--
Rol|>R--3---5---b7--
  G|------------5---
  D|--------7-------
  A|5---9-----------
```

Bar 5 - E7

Four quarter-note chord tones.

```
Nte|E---G#--B---D---
Deg|5---7---2---4---
Rol|>R--3---5---b7--
  G|------------7---
  D|----6---9-------
  A|7---------------
```

Bar 6 - F#m7b5

Four quarter-note chord tones.

```
Nte|F#--A---C---E---
Deg|6---1---b3--5---
Rol|>R--b3--b5--b7--
  G|--------5---9---
  D|----7-----------
  A|9---------------
```

Bar 7 - G#m7b5

Four quarter-note chord tones.

```
Nte|G#--B---D---F#--
Deg|7---2---4---6---
Rol|>R--b3--b5--b7--
  B|------------7---
  G|--------7-------
  D|6---9-----------
```

Bar 8 - AmMaj7

Four quarter-note chord tones.

```
Nte|A---C---E---G#--
Deg|1---b3--5---7---
Rol|>R--b3--5---7---
  B|------------9---
  G|----5---9-------
  D|7---------------
```

Scale Exercises - Section IX, Level 71

Module J - sixteen-bar mode-function capstone, bars 9-16

32 TONES | SECOND HALF

FOUR CHORD TONES PER BAR; ANNOUNCE THE MODE OR CHORD FUNCTION BEFORE BEAT 1

The first eight bars rotate through the complete parent harmony. The second half establishes functional cadences and tonic colors. Later passes may add one characteristic mode tone after the beat-one target.

Bar 9 - D7

Four quarter-note chord tones.

```
Nte|D---F#--A---C---
Deg|4---6---1---b3--
Rol|>R--3---5---b7--
  D|------------10--
  A|--------12------
  E|10--14----------
```

Bar 10 - E7

Four quarter-note chord tones.

```
Nte|E---G#--B---D---
Deg|5---7---2---4---
Rol|>R--3---5---b7--
  D|------------12--
  A|----11--14------
  E|12--------------
```

Bar 11 - AmMaj7

Four quarter-note chord tones.

```
Nte|A---C---E---G#--
Deg|1---b3--5---7---
Rol|>R--b3--5---7---
  G|------------13--
  D|----10--14------
  A|12--------------
```

Bar 12 - Am6

Four quarter-note chord tones.

```
Nte|A---C---E---F#--
Deg|1---b3--5---6---
Rol|>R--b3--5---6---
  G|------------11--
  D|----10--14------
  A|12--------------
```

Bar 13 - F#m7b5

Four quarter-note chord tones.

```
Nte|F#--A---C---E---
Deg|6---1---b3--5---
Rol|>R--b3--b5--b7--
  D|------------14--
  A|----12--15------
  E|14--------------
```

Bar 14 - Bm7

Four quarter-note chord tones.

```
Nte|B---D---F#--A---
Deg|2---4---6---1---
Rol|>R--b3--5---b7--
  G|------------14--
  D|----12--16------
  A|14--------------
```

Bar 15 - E7

Four quarter-note chord tones.

```
Nte|E---G#--B---D---
Deg|5---7---2---4---
Rol|>R--3---5---b7--
  D|------------12--
  A|--------14------
  E|12--16----------
```

Bar 16 - AmMaj7

Four quarter-note chord tones.

```
Nte|A---C---E---G#--
Deg|1---b3--5---7---
Rol|>R--b3--5---7---
  G|------------13--
  D|--------14------
  A|12--15----------
```

Final pass: bars 1-8 name the parent-key chord; bars 9-16 name the local mode or tonic-color function.

Scale Exercises - Section IX, Level 71

Practice progression, error diagnosis, mastery gate, and log

NEXT: LYDIAN DOMINANT AND ALTERED-DOMINANT SYNTHESIS

NAME THE CHORD, MODE, CHARACTERISTIC NOTE, AND DESTINATION

Ten-phase practice progression

- [ ] 1. One interval family in P1
- [ ] 2. All intervals in one position
- [ ] 3. Triad overlays
- [ ] 4. Seventh overlays
- [ ] 5. Seven mode mirrors
- [ ] 6. Mode-function cells
- [ ] 7. Five cadence phrases
- [ ] 8. Alternate progressions
- [ ] 9. D Lydian dominant routes
- [ ] 10. Resolutions and capstone

Error diagnosis

MODE WITHOUT CHORD	return to the arpeggio and drone
LOST #4	isolate D-F#-G#-A over D7
LOST b6	isolate B-C-D-E over E7
ALTERED SOUP	play G#-C-F# first; add one alteration
WRONG MODE 7 NAME	state destination: A minor or C# minor
INTERVAL FRET SEARCH	name both notes before moving
PICK RESET	air-pick the complete twelve-tone cell

Mastery gate before Level 72

- [ ] Complete all twenty interval cells and name their actual interval qualities.
- [ ] Arpeggiate all seven triads and all seven seventh chords from memory.
- [ ] Play and name all seven modes with the correct chord function.
- [ ] Perform the five iMaj7-IV7-V7-iMaj7 phrases and all three alternate progressions.
- [ ] Play D Lydian dominant in all five positions with an audible #4.
- [ ] Demonstrate both G#m7b5 -> AmMaj7 and G#7alt -> C#m7 interpretations.
- [ ] Perform the sixteen-bar capstone on two separate days without losing the mode-function labels.

Practice log

Date	Module / mode	Chord function	D/U	BPM	Clean?	Weak interval / color	Next target

Standard tuning. Fixed key: A melodic minor. Modes are tied to chord function, not treated as unrelated scale boxes.